

A fast algorithm for putting your network to bed

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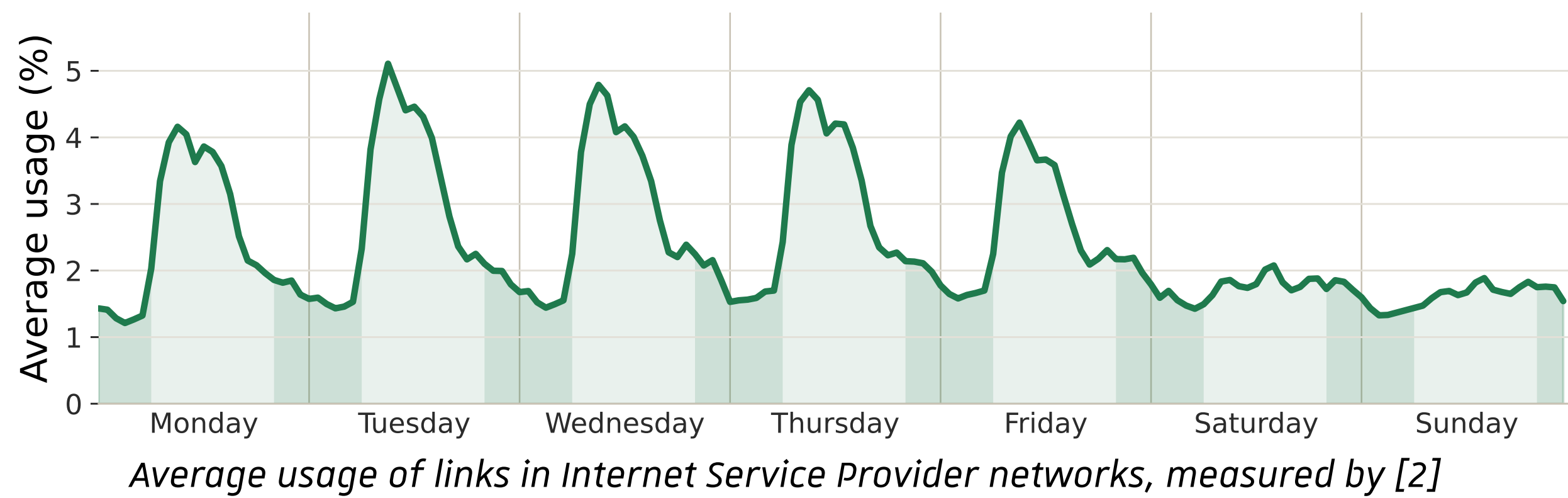
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1 - Context

- Networks **consume** a lot of energy
 - 5% of France's CO₂ production relative to digital media [1]
- Networks are **over-provisioned**, especially Internet Service Providers



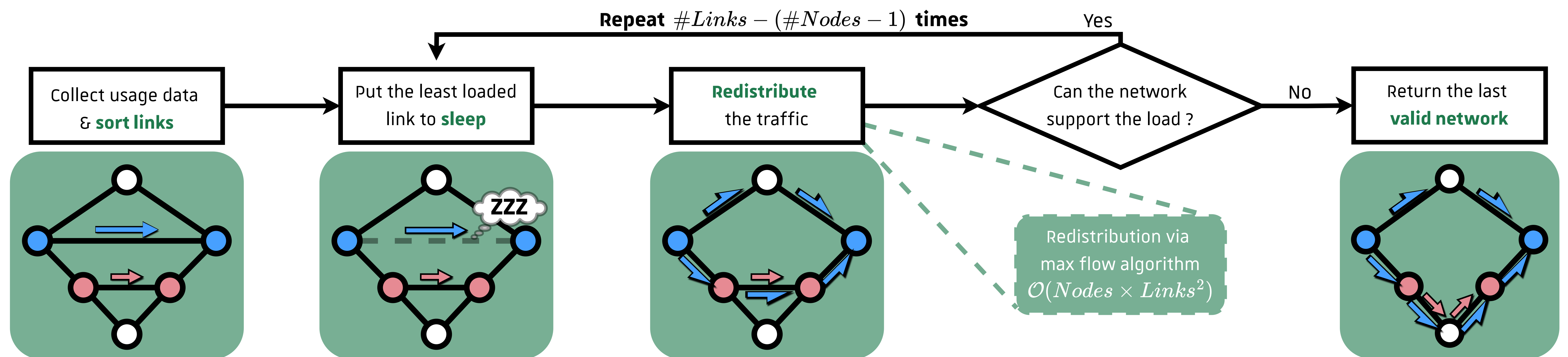
2 - Objective: Saving energy...

- By **turning off** as many **links** as possible

... while:

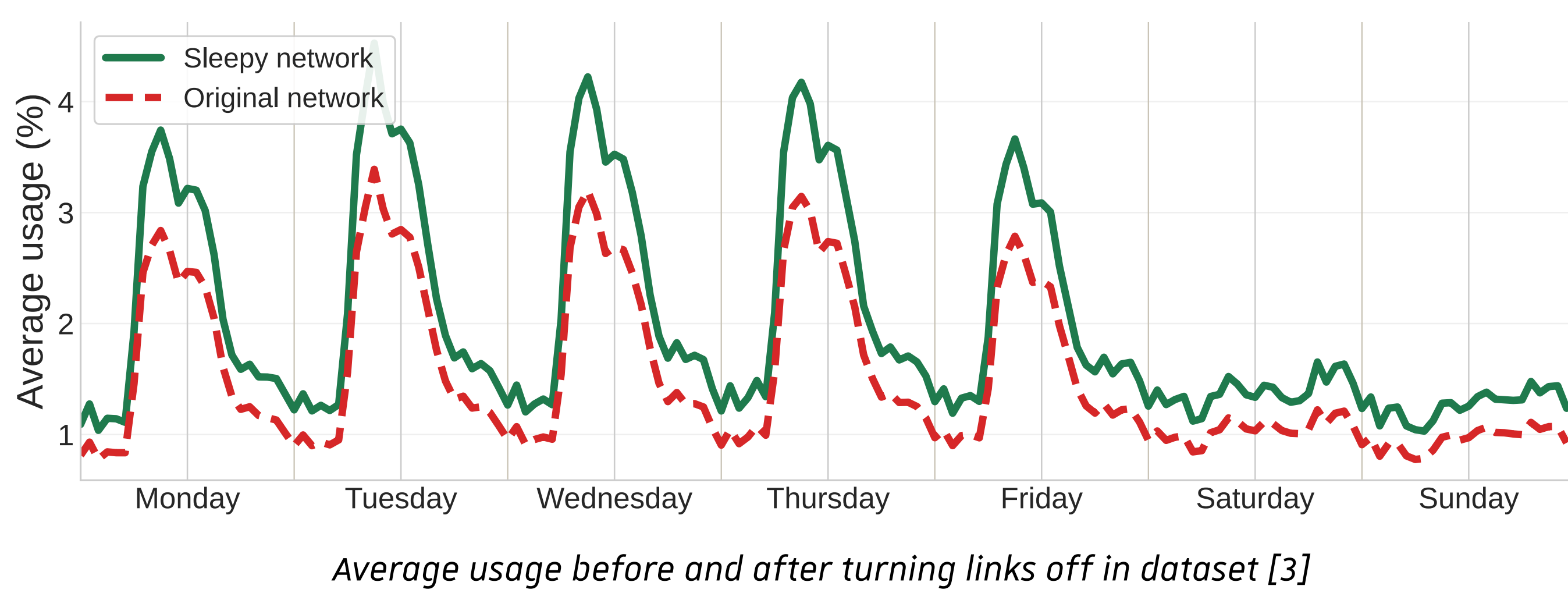
- Making sure the network can support usual traffic
- Having a **fast execution time** :
 - The problem is $\mathcal{NP}$ -Hard [2]
 - Networks are ever changing (failures, updates, ...)

3 - Algorithm: Sleep and Redistribute, an efficient $\mathcal{O}(\text{Nodes} \times \text{Links}^3)$ heuristic for an $\mathcal{NP}$ -Hard problem

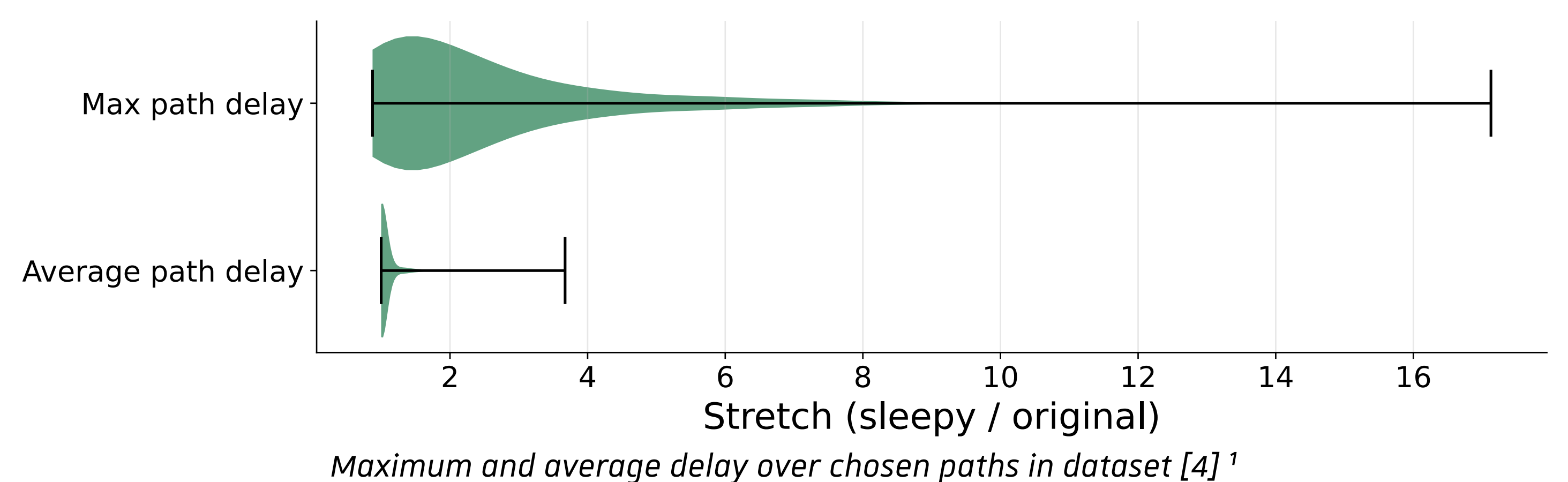


4 - Results

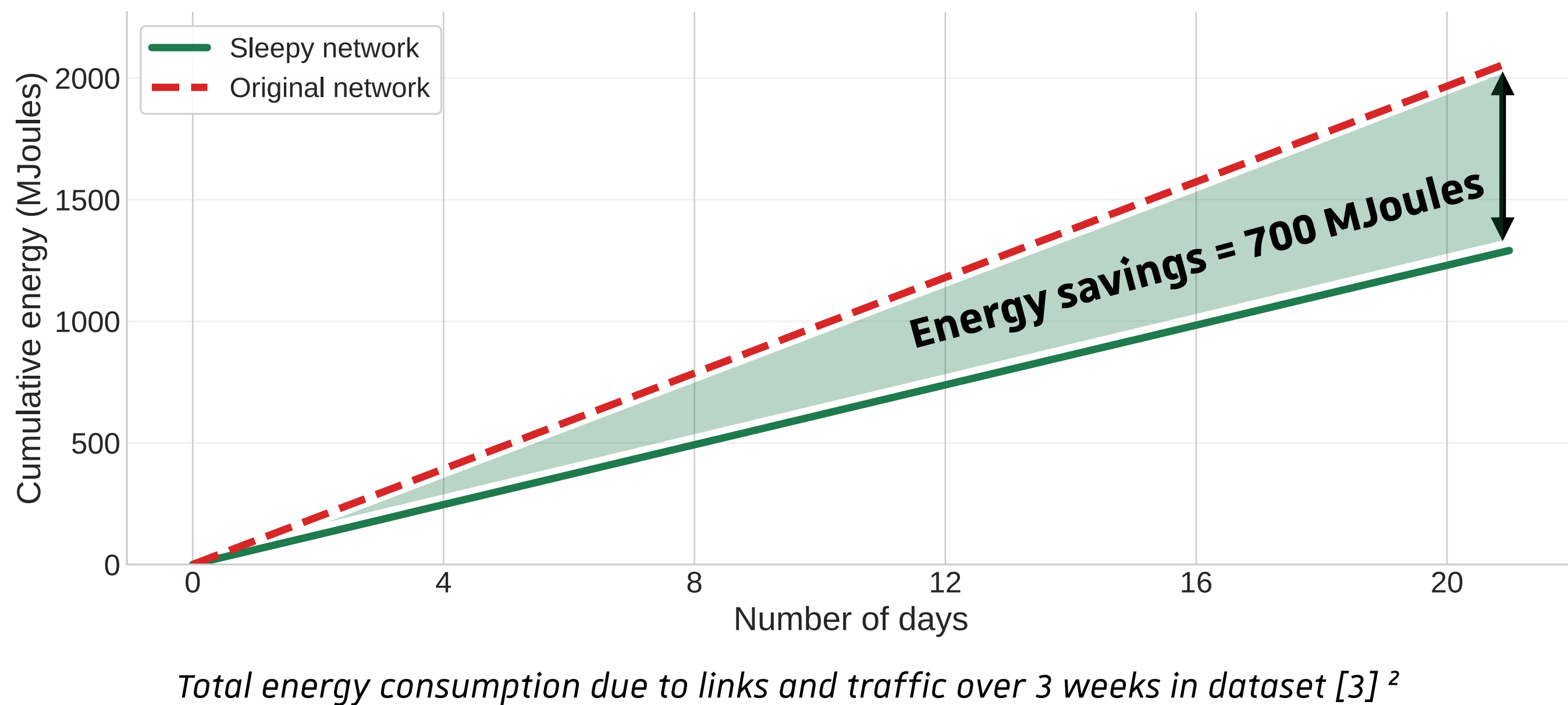
- Traffic load increases ... from 3% to 4%



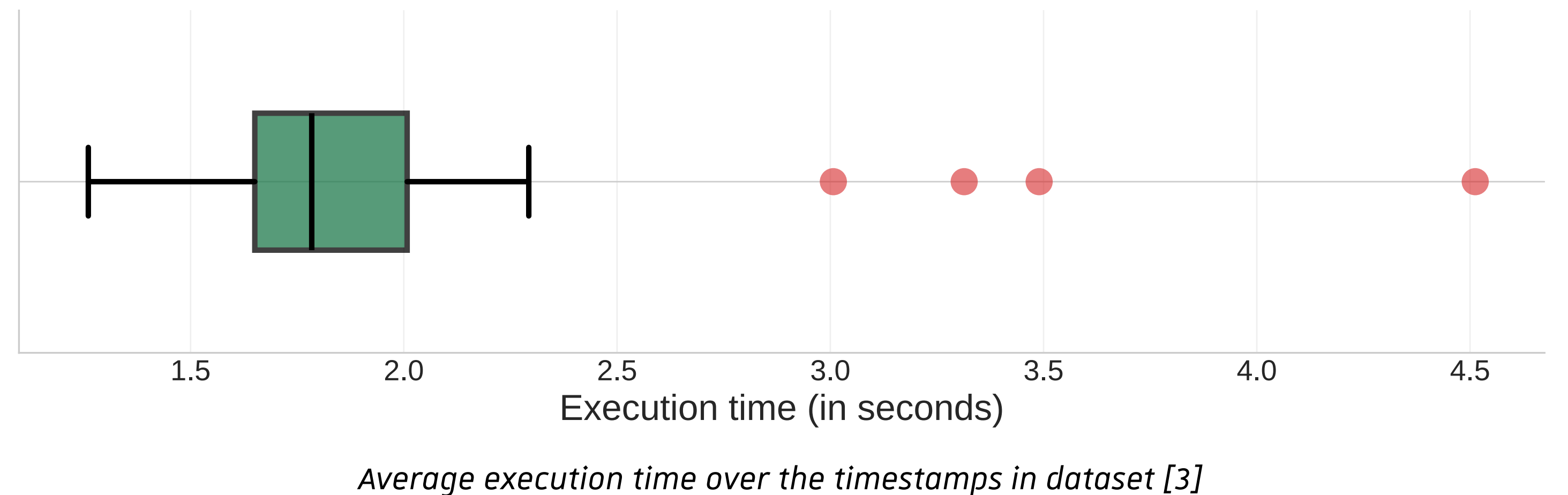
- Maximum and average delay increase ... marginally



- Links are turned off; therefore, power consumption goes down



- Computation is fast even for large networks (749 links)



Summary

- The **energy use of networks** can be reduced by turning off links
- We propose a **fast algorithm** to find links to turn off that:
 - Keeps a low average usage
 - Keeps a reasonable delay stretch

Future work

- Compensate for the lack of reliability (dynamically turn links on/off)
- Mitigate the increase of delay by adding constraints
- Experiment on a **programmable testbed**

References

- [1] : ADEME & Arcep, Evaluation de l'impact environnemental du numérique en France et analyse prospective — Analyse prospective à 2030 et 2050, 2023
- [2] : Ilsen, M., Otten, D., Aschenbruck, N., & Chimani, M. (2026). No Traffic to Cry: Traffic-Oblivious Link Deactivation for Green Traffic Engineering (Version 1). arXiv. <https://doi.org/10.48550/ARXIV.2601.13087>
- [3] : Jacob, R. (2024). ISP topology and traffic dataset (Version 1.0) [Dataset]. Zenodo. <https://doi.org/10.5281/ZENODO.12580395>
- [4] : Gay, S., Schaus, P., & Vissicchio, S. (2017). REPETITA: Repeatable Experiments for Performance Evaluation of Traffic-Engineering Algorithms (Version 1). arXiv. <https://doi.org/10.48550/ARXIV.1710.08665>

¹ : The [3] dataset does not exhibit information about delay; the [4] dataset does not exhibit information about link load. Therefore, we construct a realistic synthetic network by combining the topology and link delays from one dataset with the statistical characteristics of link loads from the other.

² : Experiments done on Intel's programmable chip: Tofino embedded in the WEDGE-100BF-32X chassis show that each link consumes about 1.5W and the processing of packets consumes about 0.025W per Gigabit / sec.